MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY TTL, QUADRUPLE BUS BUFFER GATES WITH THREE-STATE OUTPUTS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, low-power Schottky, quadruple bus buffer gates with three-state outputs. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number. Provisions for radiation hardness assurance (RHA) to four radiation test levels is included.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Quadruple bus buffer gate (inverting control input)
02	Quadruple bus buffer gate (noninverting control input)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

1.2.4 Radiation hardness level. Radiation hardness levels shall be as defined in MIL-M-38510.

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/:	
Device type 01	110 mW dc
Device type 02	121 mW dc
Lead temperature (soldering, 10 seconds)	+300°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Thermal resistance, junction-to-case (θ_{JC}):	
Cases A, C, and D	(See MIL-M-38510, appendix C)
Case 2	60°C/W 2/
Junction temperature (T_J) 3/	+175°C

1.4 Recommended operating conditions.

Low-level output current (I_{OL})	12 mA maximum.
High-level output current (I_{OH})	-1.0 mA maximum.
Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum.
Minimum high level input voltage (V_{IH})	+2.0 V maximum.
Maximum low level input voltage (V_{IL})	+0.7 V maximum.
Case operating temperature range (T_C)	-55°C to +125°C.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets of MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be in accordance with MIL-M-38510 and as specified herein.

3.2.1 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

2/ When a thermal resistance value is included in MIL-M-38510, Appendix C, it shall supersede the value stated herein.
3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

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3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and shall apply over the full recommended case operating temperature range.

3.4.1 Post-irradiation performance characteristics. The electrical performance characteristics of radiation hardness assured devices following exposure to the designated radiation levels are as specified in table III, subgroups 1 and 9 and apply at an ambient temperature of 25°C.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 9 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test conditin D, E, or F, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical tests shall be as specified in table II, except interim electrical tests prior to burn-in are optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in metnod 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.4).

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA, T_C = +25°C	01,02		-1.5	V
High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IH} = 2.0 V, V_{IL} = 0.7 V, I_{OH} = -1 mA	01,02	2.4		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 12 mA, V_{IH} = 2.0 V, V_{IL} = 0.7 V	01,02		0.4	V
Off-state (high-impedance state) output current	$I_{O(off)1}$	V_{CC} = 5.5 V, V_{OH} = 2.4 V, V_{IL} = 0.7 V, V_{IH} = 2.0 V	01,02		20	µA
Off-state (high-impedance state) output current	$I_{O(off)2}$	V_{CC} = 5.5 V, V_{OH} = 0.4 V, V_{IL} = 0.7 V, V_{IH} = 2.0 V	01,02		-20	µA
High level input current (all inputs)	I_{IH1}	V_{CC} = 5.5 V, V_{IH} = 2.7 V	01,02		20	µA
High level input current (all inputs)	I_{IH2}	V_{CC} = 5.5 V, V_{IH} = 5.5 V	01,02		100	µA
Low level input current at control input	I_{IL1}	V_{CC} = 5.5 V, V_{IL} = 0.4 V	01,02	0	-400	µA
Low level input current at data input (control high) (circuits A and B only)	I_{IL2}	V_{CC} = 5.5 V, V_{IH} = 5.5 V, V_{IL} = 0.4 V	01		-40	µA
Low level input current at data input (control low) (circuits A and B only)	I_{IL2}	V_{CC} = 5.5 V Control: V_{IL} = 0 V Data: V_{IL} = 0.4 V	02		-40	µA
Low level input current at data input (control low)	I_{IL3}	V_{CC} = 5.5 V Control: V_{IL} = 0 V Data: V_{IL} = 0.4 V	01	0	-400	µA
Low level input current at data input (control high)	I_{IL3}	V_{CC} = 5.5 V, V_{IH} = 5.5 V, V_{IL} = 0.4 V	02	0	-400	µA
Supply current	I_{CC}	V_{CC} = 5.5 V Control = 4.5 V Data = 0 V	01		20	mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
Supply current	I_{CC}	V_{CC} = 5.5 V Control = 0 V Data = 0 V	02		22	mA
Short circuit output current 2/	I_{OS}	V_{CC} = 5.5 V	01,02	-40	-225	mA
Propagation delay time (low-to-high level)	t_{PLH}	C_L = 50 pF, R_L = 110Ω V_{CC} = 5.0 V	01,02	2	20	ns
Propagation delay time (high-to-low level)	t_{PHL}	C_L = 50 pF, R_L = 110Ω V_{CC} = 5.0 V	01,02	2	24	ns
Output enable time to high level	t_{PZH}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 110Ω	01	2	28	ns
			02	2	33	ns
Output enable time to low level	t_{PZL}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 110Ω	01	2	33	ns
			02	2	46	ns
Output disable time from high level	t_{PHZ}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 110Ω	01	2	41	ns
			02	2	48	ns
Output disable time from low level	t_{PLZ}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 110Ω	01	2	33	ns
			02	2	39	ns

1/ Complete terminal conditions shall be as specified in table III.

2/ Not more than one output should be shorted at a time. The duration of any short circuit should not exceed 5 seconds.

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4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, 6, 7, and 8 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical paramters shall be as specified in table II herein.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,9, 10, 11	1*,2,3,9
Group A test requirements (method 5005)	1,2,3,9, 10, 11	1,2,3,9
Group B test requirements (method 5005) subgroup 5	1,2,3,9, 10, 11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3
Group E end-point electrical parameters (method 5005)	1, 9	1, 9

*PDA applies to subgroup 1 (see 4.2c).

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical test parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

c. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D, E, or F using the circuit shown on figure 3, or equivalent.

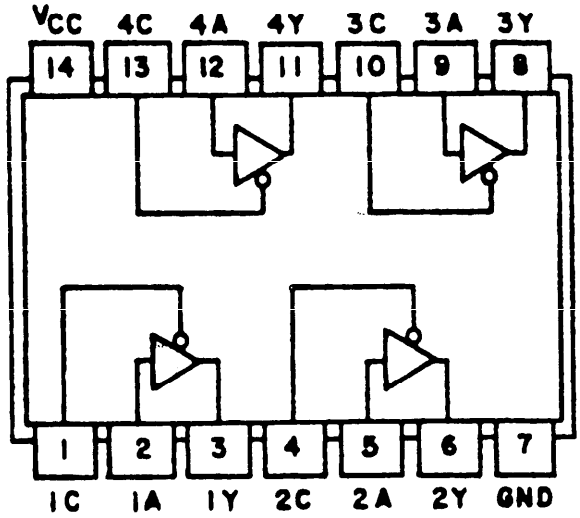
(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

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Device type 01

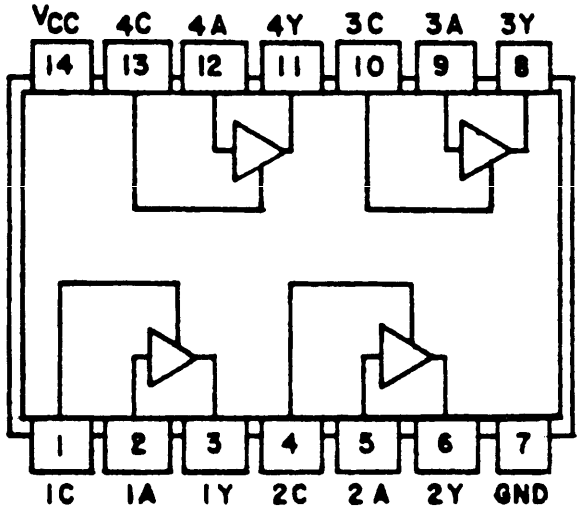
Case C and D



Positive logic:
Y = A
Output is off (disabled) when C is high.

Device type 02

Case C and D

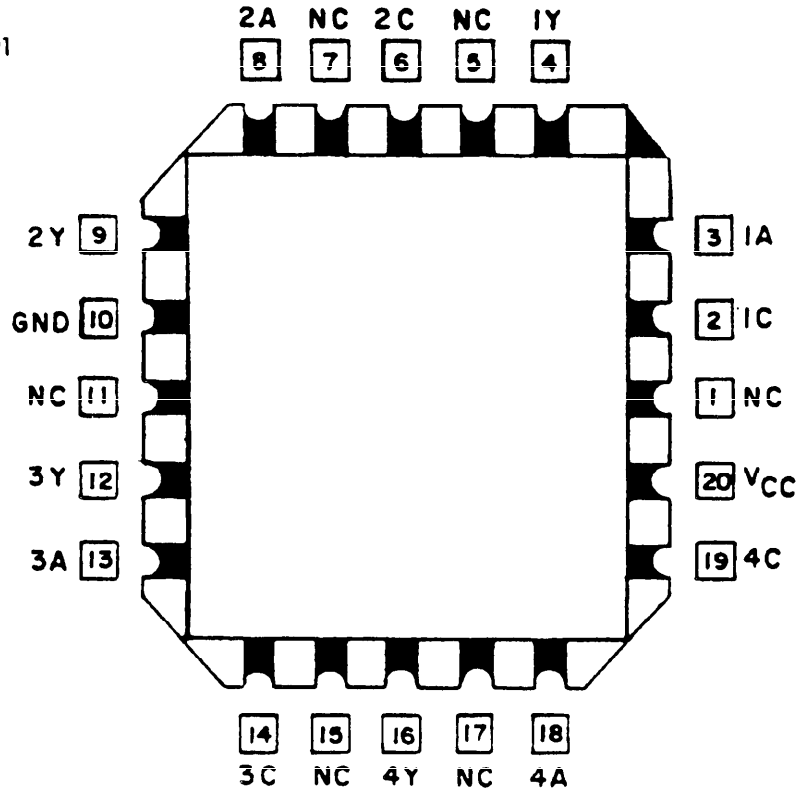


Positive logic:
Y = A
Output is off (disabled) when C is low.

FIGURE 1. Logic diagram and terminal connections.

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Device type 01
Case 2



Device type 02
Case 2

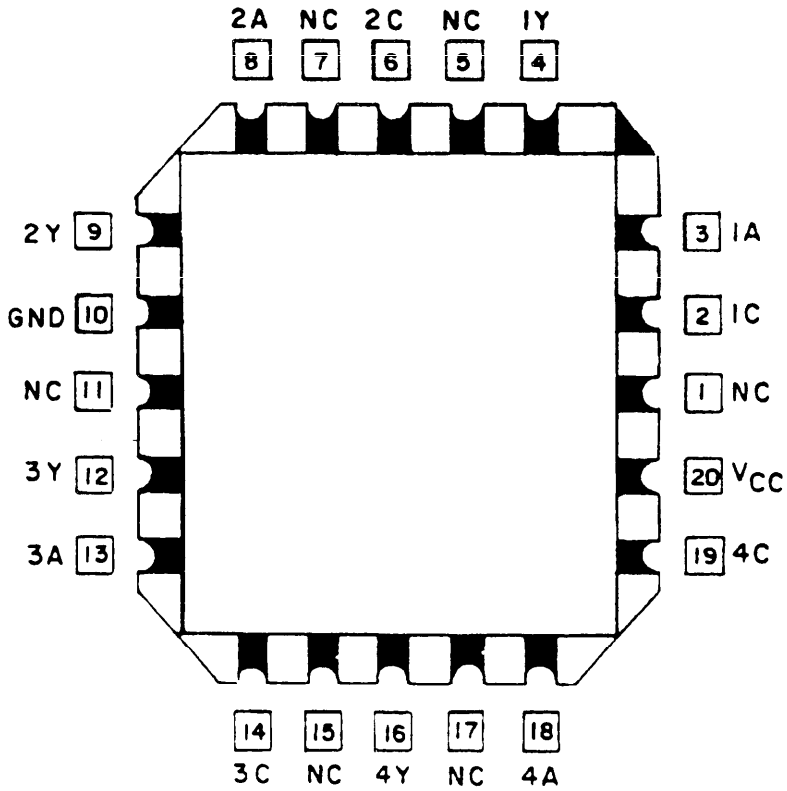


FIGURE 1. Logic diagram and terminal connections - Continued.

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Device type 01

INPUTS		OUTPUT
E	D	
L	L	L
L	H	H
H	X	Z

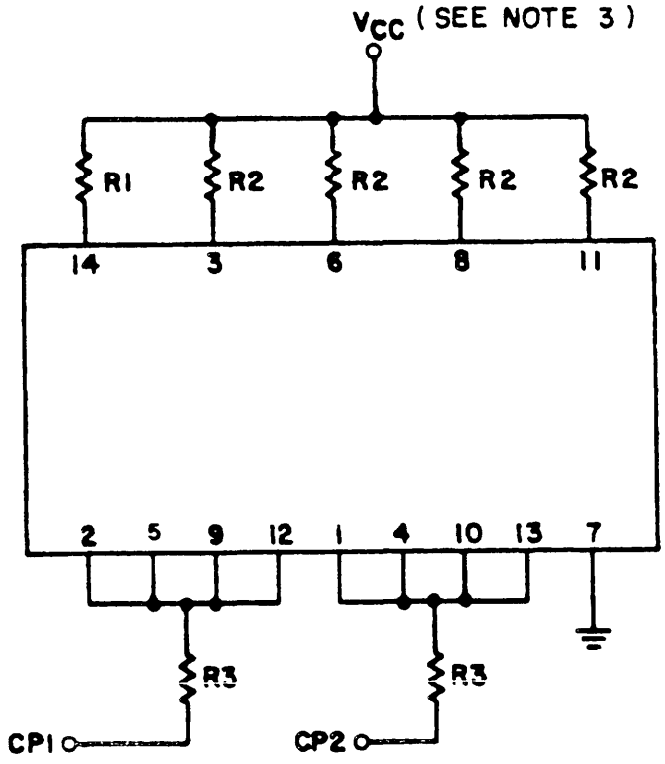
Device type 02

INPUTS		OUTPUT
E	D	
H	L	L
H	H	H
L	X	Z

H = High level logic
L = Low level logic
X = Doesn't matter
Z = High impedence (off)

FIGURE 2. Truth tables.

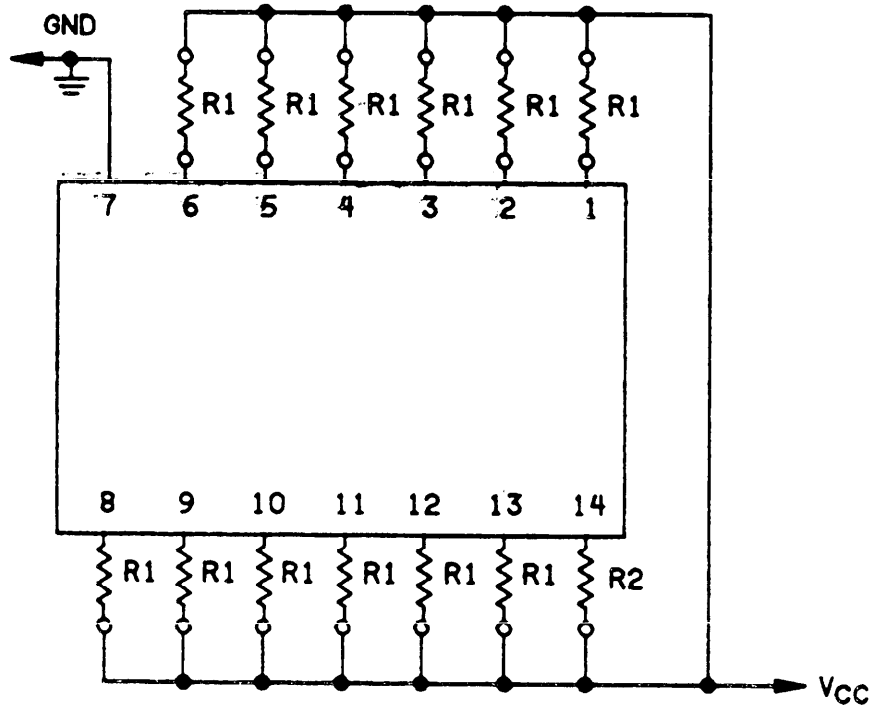
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NOTES:
1. CP1 = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. CP2 = 50 kHz ±50% square wave, otherwise as described in note 1 above.
3. R_1 and V_{CC} shall be chosen so as to insure 5.0 V at the device terminal.
4. R_2 = 470Ω ±5%, R_3 = 27Ω ±5%.

FIGURE 3. Burn-in and life-test circuit for device types 01 and 02.

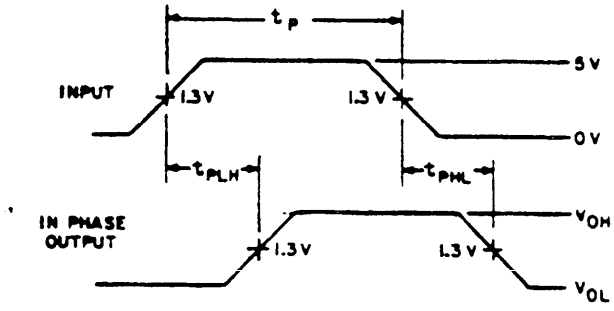
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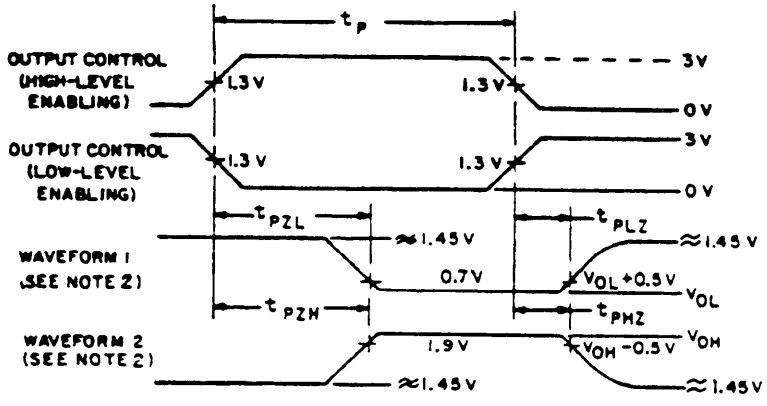
NOTES:
1. High temperature (175°C) condition F burn-in circuit only.
2. R1 = 2k Ω, R2 = 2Ω.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

FIGURE 3. Burn-in and life-test circuit for device types 01 and 02 - Continued.

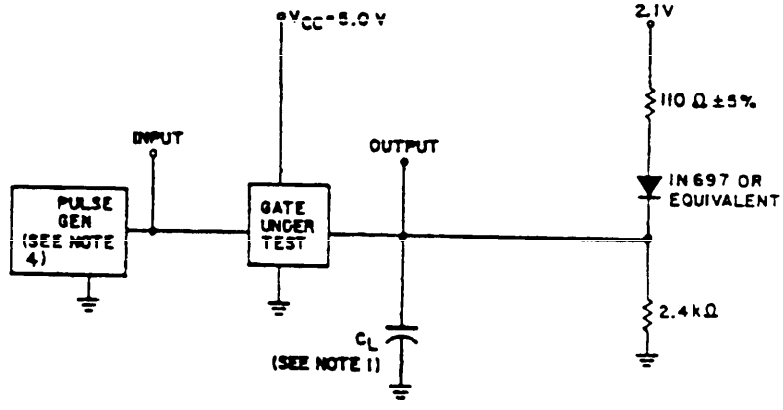
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VOLTAGE WAVEFORMS
PROPAGATION DELAY TIMES



VOLTAGE WAVEFORMS
ENABLE AND DISABLE TIMES,
THREE - STATE OUTPUTS

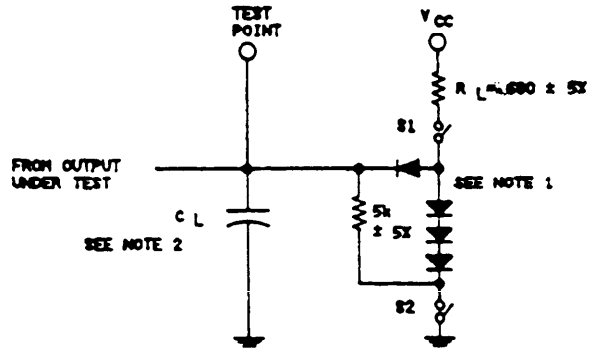


Test circuit

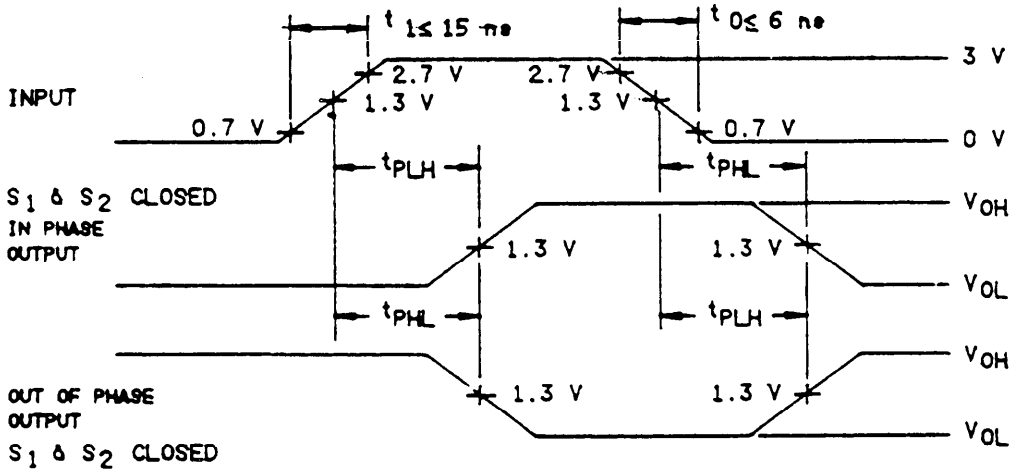
NOTES:
1. C_L = 50 pF ±10% minimum for all tests.
   C_L includes scope probe and jig capacitance.
2. Waveform 1 is for an output with internal conditions such that the output is low except when disabled by the output control.
   Waveform 2 is for an output with internal conditions such that the output is high except when disabled by the output control.
3. In the examples above, the phase relationships between inputs and outputs have been chosen arbitrarily.
4. All input pulses are supplied by generators having the following characteristics: PRR ≤ 1 MHz, t_p = 500 ns, $Z_{out} \approx 50\Omega$, V_{gen} = 3.0 V, and $t_r \leq 15$ ns, $t_f \leq 6$ ns between 0.7 V and 2.7 V.

FIGURE 4. Switching time test circuit and waveforms.

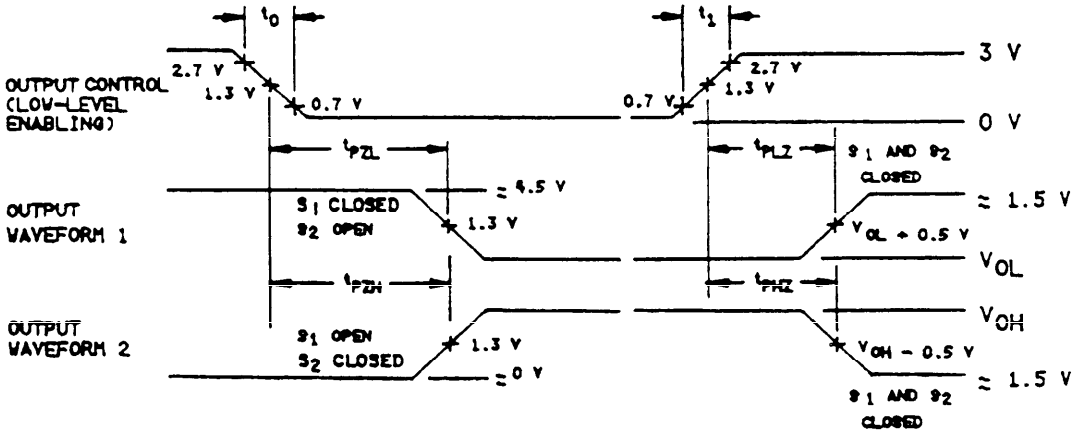
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ALTERNATE LOAD CIRCUIT



ALTERNATE VOLTAGE WAVEFORMS
PROPAGATION DELAY TIMES



ALTERNATE VOLTAGE WAVEFORMS
FOR TRI-STATE SWITCHING

NOTES:
1. Diodes are 1N3064 or equivalent.
2. C_L = 50 pF ±10% for t_{PLH}, t_{PHL}, t_{PZL} and t_{PZH}; C_L = 15 pF minimum for t_{PHZ} and t_{PLZ}. C_L includes probe and jig capacitance.

FIGURE 4. Switching time test circuit and waveforms - Continued.

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Device type 01

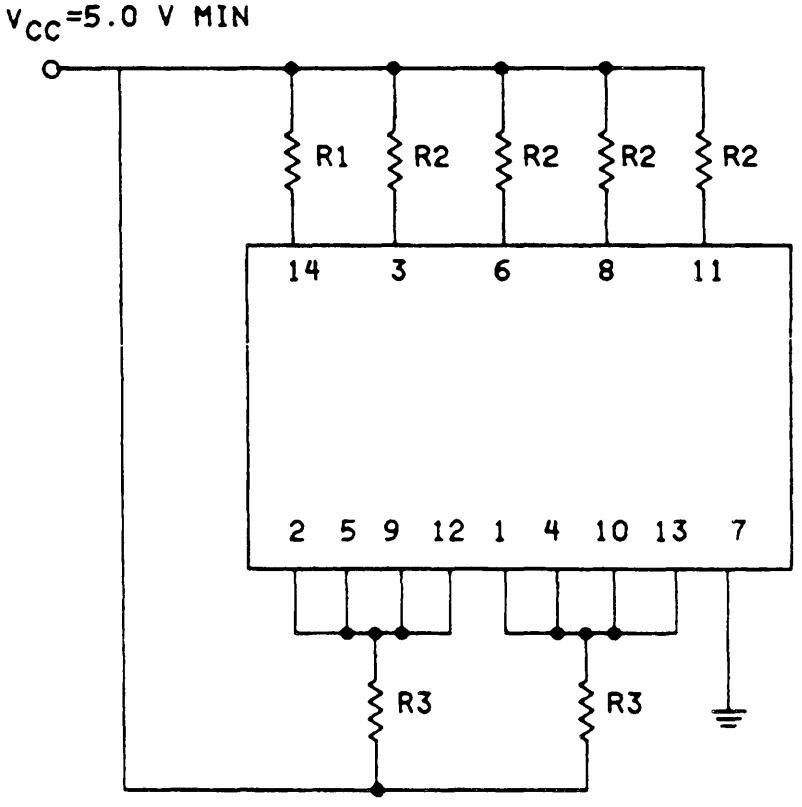
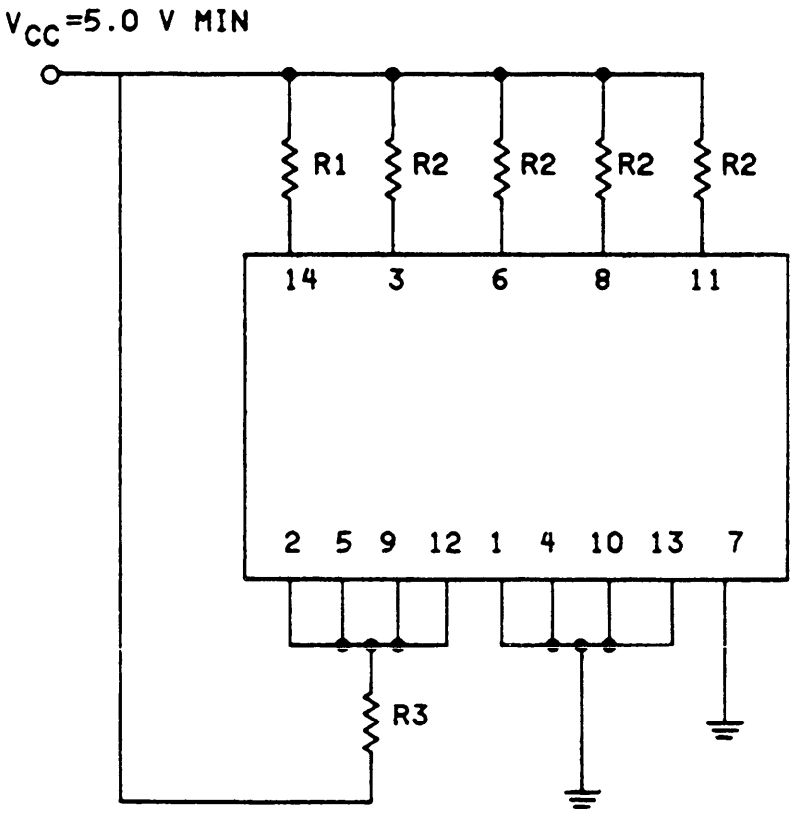


FIGURE 5. Bias condition for total dose radiation test.

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Device type 02



Device type	V_{CC} (volts)*	R_1 (ohms)	R_2 (ohms)	R_3 (ohms)
01	5.5	0	1.3k	1.3k
02	5.5	0	1.3k	1.3k

*V_{CC} shall be high enough to insure that a 5.0 volt minimum is present at device V_{CC} terminal.

FIGURE 5. Bias condition for total dose radiation test - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	1C	1A	1Y	2C	2A	2Y	GND	3Y	3A	3C	4Y	4A	4C	V_{CC}				
T_C = +25°C	V_{IC}		1	4.5 V	-18 mA					GND							4.5 V	1A		-1.5	V
			2	-18 mA						•							•	1C		•	•
			3				4.5 V	-18 mA		•							•	2A		•	•
			4				-18 mA			•							•	2C		•	•
			5							•		-18 mA	4.5 V				•	3A		•	•
			6							•			-18 mA				•	3C		•	•
			7							•					-18 mA	4.5 V	•	4A		•	•
			8							•						-18 mA	•	4C		•	•
	V_{OH}	3006	9	0.7 V	2.0 V	-1 mA				•							•	1Y	2.4		•
		"	10				0.7 V	2.0 V	-1 mA	•							•	2Y	•		•
		"	11							•	-1 mA	2.0 V	0.7 V				•	3Y	•		•
		"	12							•				-1 mA	2.0 V	0.7 V	•	4Y			•
	V_{OL}	3007	13	0.7 V	0.7 V	12 mA				•							•	1Y		0.4	•
		"	14				0.7 V	0.7 V	12 mA	•							•	2Y		•	•
		"	15							•	12 mA	0.7 V	0.7 V				•	3Y		•	•
		"	16							•				12 mA	0.7 V	0.7 V	•	4Y			•
	$I_{O(off)1}$		17	2.0 V	0.7 V	2.4 V				•							5.5 V	1Y		20	μA
			18				2.0 V	0.7 V	2.4 V	•							•	2Y		•	•
			19							•	2.4 V	0.7 V	2.0 V				•	3Y		•	•
			20							•				2.4 V	0.7 V	2.0 V	•	4Y			•
	$I_{O(off)2}$		21	2.0 V	2.0 V	0.4 V				•							•	1Y		-20	•
			22				2.0 V	2.0 V	0.4 V	•							•	2Y		•	•
			23							•	0.4 V	2.0 V	2.0 V				•	3Y		•	•
			24							•				0.4 V	2.0 V	2.0 V	•	4Y			•
	I_{IH1}	3010	25	2.7 V						•							•	1C		20	•
		"	26	GND	2.7 V					•							•	1A		•	•
		"	27				2.7 V	2.7 V		•							•	2C		•	•
		"	28				GND			•							•	2A		•	•
		"	29							•		2.7 V	GND				•	3A		•	•
		"	30							•			2.7 V				•	3C		•	•
		"	31							•					2.7 V	GND	•	4A		•	•
		"	32							•						2.7 V	•	4C		•	•
	I_{IH2}	"	33	5.5 V						•							•	1C		100	•
		"	34	GND	5.5 V					•							•	1A		•	•
		"	35				5.5 V	5.5 V		•							•	2C		•	•
		"	36				GND			•							•	2A		•	•
		"	37							•		5.5 V	GND				•	3A		•	•
		"	38							•			5.5 V				•	3C		•	•
		"	39							•					5.5 V	GND	•	4A		•	•
		"	40							•						5.5 V	•	4C		•	•
	I_{IL1}	3009	41	0.4 V						•							•	1C	3/	3/	•
		"	42				0.4 V			•							•	2C	•	•	•
		"	43							•			0.4 V				•	3C		•	•
		"	44							•						0.4 V	•	4C		•	•
	I_{IL2} CRT A,B	"	45	5.5 V	0.4 V					•							•	1A		-40	•
		"	46				5.5 V	0.4 V		•							•	2A		•	•
		"	47							•		0.4 V	5.5 V				•	3A		•	•
		"	48							•					0.4 V	5.5 V	•	4A		•	•

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,C,D / Case 2 / Test no.	1 / 2 / 1C	2 / 3 / 1A	3 / 4 / 1Y	4 / 6 / 2C	5 / 8 / 2A	6 / 9 / 2Y	7 / 10 / GND	8 / 12 / 3Y	9 / 13 / 3A	10 / 14 / 3C	11 / 16 / 4Y	12 / 18 / 4A	13 / 19 / 4C	14 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_{IL3}	3009	49 50 51 52	2/	0.4 V		2/	0.4 V		GND		0.4 V	2/		0.4 V	2/	5.5 V	1A 2A 3A 4A	3/	3/	µA
	I_{CC}	3005	53	4.5 V	GND		4.5 V	GND				GND	4.5 V		GND	4.5 V		V_{CC}		20	mA
	I_{OS} 4/	3011	54 55 56 57	GND	4.5 V	GND	GND	4.5 V	GND		GND	4.5 V	GND	GND	4.5 V	GND		1Y 2Y 3Y 4Y	-40	-225	
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C; and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = +25°C	t_{PHL}	3003 Fig. 4	58 59 60 61	GND	IN	OUT	GND	IN	OUT	GND	OUT	IN	GND	OUT	IH	GND	5.0 V	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y	2	18	ns
	t_{PLH}		62 63 64 65	GND	IN	OUT	GND	IN	OUT		OUT	IN	GND	OUT	IH	GND		1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y		15	
	t_{PZH}		66 67 68 69	IN	5.0 V	OUT	IN	5.0 V	OUT		OUT	5.0 V	IN	OUT	5.0 V	IN		1C to 1Y 2C to 2Y 3C to 3Y 4C to 4Y		20	
	t_{PZL}		70 71 72 73	IN	GND	OUT	IN	GND	OUT		OUT	GND	IN	OUT	GND	IN		1C to 1Y 2C to 2Y 3C to 3Y 4C to 4Y		28	
	t_{PHZ}		74 75 76 77	IN	5.0 V	OUT	IN	5.0 V	OUT		OUT	5.0 V	IN	OUT	5.0 V	IN		1C to 1Y 2C to 2Y 3C to 3Y 4C to 4Y		37	
	t_{PLZ}		78 79 80 81	IN	GND	OUT	IN	GND	OUT		OUT	GND	IN	OUT	GND	IN		1C to 1Y 2C to 2Y 3C to 3Y 4C to 4Y		25	

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,C,D Case 2 / Test no.	1 / 1C	2 / 1A	3 / 1Y	4 / 2C	5 / 2A	6 / 2Y	7 / GND	8 / 3Y	9 / 3A	10 / 3C	11 / 4Y	12 / 4A	13 / 4C	14 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
10 T_C = +125°C	t_{PHL}	3003 Fig. 4	82	GND	IN	OUT	GND	IN	OUT	GND	OUT	IN	GND	OUT	IN	GND	5.0 V	1A to 1Y	2	24	ns
			83															2A to 2Y			
			84															3A to 3Y			
			85															4A to 4Y			
	t_{PLH}		86	GND	IN	OUT	GND	IN	OUT		OUT	IN	GND	OUT	IN	GND		1A to 1Y		20	
			87															2A to 2Y			
			88															3A to 3Y			
			89															4A to 4Y			
	t_{PZH}		90	IN	5.0 V	OUT	IN	5.0 V	OUT		OUT	5.0 V	IN	OUT	5.0 V	IN		1C to 1Y		28	
			91															2C to 2Y			
			92															3C to 3Y			
			93															4C to 4Y			
	t_{PZL}		94	IN	GND	OUT	IN	GND	OUT		OUT	GND	IN	OUT	GND	IN		1C to 1Y		33	
			95															2C to 2Y			
			96															3C to 3Y			
			97															4C to 4Y			
	t_{PHZ}		98	IN	5.0 V	OUT	IN	5.0 V	OUT		OUT	5.0 V	IN	OUT	5.0 V	IN		1C to 1Y		41	
			99															2C to 2Y			
			100															3C to 3Y			
			101															4C to 4Y			
	t_{PLZ}		102	IN	GND	OUT	IN	GND	OUT		OUT	GND	IN	OUT	GND	IN		1C to 1Y		33	
			103															2C to 2Y			
			104															3C to 3Y			
			105															4C to 4Y			
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																				

1/ Pins not referenced are NC.

2/ For circuits A and B, pin conditions shall be low ≤0.7 V; for circuits C, D, and E, pin conditions shall be low ≤0.7 V, high ≥2.0 V, or open.

3/ Minimum and maximum test limits for I_{IL1} and I_{IL3} shall be as follows:

Tests	Circuits A	B	C	D	E
I_{IL1}	-1.0/-150	-120/-360	0/-200	-160/-400	-140/-370
I_{IL3}	-150/-380	-160/-400	0/-200	-160/-400	-140/-370

4/ Circuits D and E tests 54 through 57, I_{OS}, min -30/max -130 mA.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,C,D / Case 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits Min	Limits Max	Unit
			Test no.	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
				1C	1A	1Y	2C	2A	2Y	GND	3Y	3A	3C	4Y	4A	4C	V_{CC}				
T_C = +25°C	V_{IC}		1	4.5 V	-18 mA					GND							4.5 V	1A		-1.5	V
			2	-18 mA														1C			
			3				4.5 V	-18 mA										2A			
			4				-18 mA											2C			
			5									-18 mA	4.5 V					3A			
			6										-18 mA					3C			
			7												-18 mA	4.6 V		4A			
			8													-18 mA		4C			
	V_{OH}	3006	9	2.0 V	2.0 V	-1 mA												1Y	2.4		
			10				2.0 V	2.0 V	-1 mA									2Y			
			11								-1 mA	2.0 V	2.0 V					3Y			
			12											-1 mA	2.0 V	2.0 V		4Y			
	V_{OL}	3007	13	2.0 V	0.7 V	12 mA												1Y		0.4	
			14				2.0 V	0.7 V	12 mA									2Y			
			15								12 mA	0.7 V	2.0 V					3Y			
			16											12 mA	0.7 V	2.0 V		4Y			
	$I_{O(off)1}$		17	0.7 V	0.7 V	2.4 V											5.5 V	1Y		20	µA
			18				0.7 V	0.7 V	2.4 V									2Y			
			19								2.4 V	0.7 V	0.7 V					3Y			
			20											2.4 V	0.7 V	0.7 V		4Y			
	$I_{O(off)2}$		21	0.7 V	2.0 V	0.4 V												1Y		-20	
			22				0.7 V	2.0 V	0.4 V									2Y			
			23								0.4 V	2.0 V	0.7 V					3Y			
			24											0.4 V	2.0 V	0.7 V		4Y			
	I_{IH1}	3010	25	2.7 V														1C		20	
			26	5.5 V	2.7 V													1A			
			27				2.7 V											2C			
			28				5.5 V	2.7 V										2A			
			29									2.7 V	5.5 V					3A			
			30										2.7 V					3C			
			31												2.7 V	5.5 V		4A			
			32													2.7 V		4C			
	I_{IH2}		33	5.5 V														1C		100	
			34	5.5 V	5.5 V													1A			
			35				5.5 V											2C			
			36				5.5 V	5.5 V										2A			
			37									5.5 V	5.5 V					3A			
			38										5.5 V					3C			
			39												5.5 V	5.5 V		4A			
			40													5.5 V		4C			
	I_{IL1}	3009	41	0.4 V														1C	1/	1/	
			42				0.4 V											2C			
			43										0.4 V					3C			
			44													0.4 V		4C			
	I_{IL2} CRT A,B		45	GND	0.4 V													1A		-40	
			46				GND	0.4 V										2A			
			47									0.4 V	GND					3A			
			48												0.4 V	GND		4A			

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,C,D Case 2 / Test no.	1 / 2 / IC	2 / 3 / 1A	3 / 4 / 1Y	4 / 5 / 2C	5 / 6 / 2A	6 / 7 / 2Y	7 / 10 / GND	8 / 12 / 3Y	9 / 13 / 3A	10 / 14 / 3C	11 / 16 / 4Y	12 / 18 / 4A	13 / 19 / 4C	14 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_{IL3}	3009	49	2/	0.4 V		2/	0.4 V		GND							5.5 V	1A	3/	3/	µA
		"	50							"							"	2A	"	"	"
		"	51							"		0.4 V	2/				"	3A	"	"	"
		"	52							"					0.4 V	2/	"	4A	"	"	"
	I_{CC}	3005	53	GND	GND		GND	GND		"		GND	GND		GND	GND	"	V_{CC}		22	mA
	I_{OS}	3011	54	4.5 V	4.5 V	GND				"							"	1Y	-40	-225	"
		"	55				4.5 V	4.5 V	GND	"							"	2Y	"	"	"
		"	56							"	GND	4.5 V	4.5 V				"	3Y	"	"	"
		"	57							"				GND	4.6 V	4.5 V	"	4Y	"	"	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C; and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9 T_C = +25°C	t_{PHL}	3003 Fig. 4	58	5.0 V	IN	OUT				GND							5.0 V	1A to 1Y	2	18	ns
		"	59				5.0 V	IN	OUT	"							"	2A to 2Y	"	"	"
		"	60							"	OUT	IN	5.0 V				"	3A to 3Y	"	"	"
		"	61							"				OUT	IN	5.0 V	"	4A to 4Y	"	"	"
	t_{PLH}	"	62	5.0 V	IN	OUT				"							"	1A to 1Y	"	15	"
		"	63				5.0 V	IN	OUT	"							"	2A to 2Y	"	"	"
		"	64							"	OUT	IN	5.0 V				"	3A to 3Y	"	"	"
		"	65							"				OUT	IN	5.0 V	"	4A to 4Y	"	"	"
	t_{PZH}	"	66	IN	5.0 V	OUT				"							"	1C to 1Y	"	25	"
		"	67				IN	5.0 V	OUT	"							"	2C to 2Y	"	"	"
		"	68							"	OUT	5.0 V	IN				"	3C to 3Y	"	"	"
		"	69							"				OUT	5.0 V	IN	"	4C to 4Y	"	"	"
	t_{PZL}	"	70	IN	GND	OUT				"							"	1C to 1Y	"	35	"
		"	71				IN	GND	OUT	"							"	2C to 2Y	"	"	"
		"	72							"	OUT	GND	IN				"	3C to 3Y	"	"	"
		"	73							"				OUT	GND	IN	"	4C to 4Y	"	"	"
	t_{PHZ}	"	74	IN	5.0 V	OUT				"							"	1C to 1Y	"	42	"
		"	75				IN	5.0 V	OUT	"							"	2C to 2Y	"	"	"
		"	76							"	OUT	5.0 V	IN				"	3C to 3Y	"	"	"
		"	77							"				OUT	5.0 V	IN	"	4C to 4Y	"	"	"
	t_{PLZ}	"	78	IN	GND	OUT				"							"	1C to 1Y	"	30	"
		"	79				IN	GND	OUT	"							"	2C to 2Y	"	"	"
		"	80							"	OUT	GND	IN				"	3C to 3Y	"	"	"
		"	81							"				OUT	GND	IN	"	4C to 4Y	"	"	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,C,D / Case Z	1 / 2	2 / 3	3 / 4	4 / 6	5 / 8	6 / 9	7 / 10	8 / 12	9 / 13	10 / 14	11 / 16	12 / 18	13 / 19	14 / 20	Measured terminal	Limits Min	Limits Max	Unit
			Test no.	1C	1A	1Y	2C	2A	2Y	GND	3Y	3A	3C	4Y	4A	4C	V_{CC}				
10 T_C = +125°C	t_{PHL}	3003 Fig. 4	82	5.0 V	IN	OUT	5.0 V	IN	OUT	GND	OUT	IN	5.0 V	OUT	IN	5.0 V	5.0 V	1A to 1Y	2	24	ns
			83															2A to 2Y			
			84															3A to 3Y			
			85															4A to 4Y			
	t_{PLH}		86	5.0 V	IN	OUT	5.0 V	IN	OUT		OUT	IN	5.0 V	OUT	IN	5.0 V		1A to 1Y		20	
			87															2A to 2Y			
			88															3A to 3Y			
			89															4A to 4Y			
	t_{PZH}		90	IN	5.0 V	OUT	IN	5.0 V	OUT		OUT	5.0 V	IN	OUT	5.0 V	IN		1C to 1Y		33	
			91															2C to 2Y			
			92															3C to 3Y			
			93															4C to 4Y			
	t_{PZL}		94	IN	GND	OUT	IN	GND	OUT		OUT	GND	IN	OUT	GND	IN		1C to 1Y		46	
			95															2C to 2Y			
			96															3C to 3Y			
			97															4C to 4Y			
	t_{PHZ}		98	IN	5.0 V	OUT	IN	5.0 V	OUT		OUT	5.0 V	IN	OUT	5.0 V	IN		1C to 1Y		48	
			99															2C to 2Y			
			100															3C to 3Y			
			101															4C to 4Y			
	t_{PLZ}		102	IN	GND	OUT	IN	GND	OUT		OUT	GND	IN	OUT	GND	IN		1C to 1Y		39	
			103															2C to 2Y			
			104															3C to 3Y			
			105															4C to 4Y			
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																				

1/ Pins not referenced are NC.

2/ For circuits A and B only, pin conditions shall be high ≥2.0 or open.

3/ Minimum and maximum test limits for I_{IL1} and I_{IL3} shall be as follows:

Tests	Circuits A	B	C	D	E
I_{IL1}	-150/-380	-120/-360	0/-200	-160/-400	-140/-370
I_{IL3}	-150/-380	-160/-400	0/-200	-160/-400	-140/-370

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4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883 End-point electrical parameters shall be as specified in table II.

4.4.5 Group E inspection. Group E inspection shall be in accordance with table V of method 5005 or MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified table II herein.

b. Radiation hardness assurance exposure (method 1019 of MIL-STD-883) conditions. Total dose exposure circuit is shown on figure 5.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special lead lengths, or lead forming. If applicable, these requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - -	Ground zero voltage potential.
V_{IN} - - - - - - - - - - -	Voltage level at an input terminal.
I_{IN} - - - - - - - - - - -	Current flowing into an input terminal.
t_{PHZ} - - - - - - - - - - -	Output disable time from high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined high level to a high-impedance (off) state.

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t_{PLZ} - - - - - - - - - - Output disable time from low level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined low level to a high-impedance (off) state.

t_{PZH} - - - - - - - - - - Output enable time to high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined high level.

t_{PZL} - - - - - - - - - - Output enable time to low level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined low level.

6.4 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54LS125A
02	54LS126

6.7 Manufacturers' designations. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuit				
	A	B	C	D	E
	Texas Instruments	Signetics Corp.	National Semiconductor Corp.	Motorola Inc.	Fairchild Semiconductor
01	X	X	X	X	X
02	X	X	X	X	X

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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1.1 Scope. This specification covers the detail requirements for monolithic

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL_M_-38510/323C	MICROCIRCUITS DIGITAL BIPOLAR LAW POWER SCHOTTKY TTL QUADRUPLE BUS BUFFER GATES W/ 3 STATE OUTPUTS MONOLITHIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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